



Manifold for T1D – nPOD Tissue-Biology Integration

Pancreatic-tissue multi-omic causal inference across T1D disease stages

nPOD is the only resource producing **pancreatic tissue across the full T1D disease spectrum** — at-risk, new-onset, established, long-duration — paired with donor clinical history and cross-modality multi-omic characterization (histology, single-cell and spatial transcriptomics, proteomics). The analytic frontier is structural: tissue-level causal inference across disease-stage strata, cross-modality integration onto a single directed graph, and a standardized fragility layer for cross-investigator comparison. **Manifold** configures causal discovery on the nPOD multi-modal substrate, criticality estimators (persistent homology on tissue manifolds, HTE learners, change-point across strata, spatial-transcriptomic coupling) feeding a composite fragility score (ΩF), and Pearl do-calculus with interdiction for tissue-level target prioritization.

What drives progression at the tissue level — from autoantibody positivity to clinical T1D to established hypoinsulinemia — is no longer a descriptive problem but a causal-inference problem on the integrated nPOD substrate.

ΩF PILLAR MAPPING – PANCREATIC TISSUE PHYSIOLOGY

ΩF PILLAR	NPOD TISSUE READING
I — Interaction	Autoantibody × HLA × tissue-feature interactions; immune infiltrate × residual β -cell phenotype coupling
R — Reserve	Residual islet mass by histology; β -cell functional-marker density; residual insulin / C-peptide immunoreactivity across strata
J — Junction	Insulitis intensity and immune-cell infiltrate at islet boundary; endocrine-exocrine interface signatures
C — Cascade	β -cell senescence and dedifferentiation cascades in situ; ER-stress and apoptosis markers; epitope-spreading signatures
T — Temporal	Disease-stage progression hazard; time-to-established-T1D from autoantibody-positive state given tissue-level covariates

PROPOSED PARTNERSHIP

Partnership with a University of Florida Diabetes Institute principal investigator affiliated with nPOD is sought for:

1. Retrospective nPOD cross-modality tissue data access under the nPOD Data Access Committee framework.
2. Co-authorship on the tissue-level causal-integration methodological manuscript, positioned as an nPOD-affiliated consortium deliverable (fourteen to sixteen weeks from data access).
3. Co-application to NIH NIDDK, Breakthrough T1D, or the Helmsley Charitable Trust for the Phase 2 arm extending integration across additional modalities and disease-stage strata.

Apex Analytica contributes the causal-inference engine, the ΩF framework, and the engineering team. Full brief: M-T1D-02-NPOD .